Junseo Lee

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Research Interests

Systems, ML for Systems, Compilers & Runtime, Hardware-Aware Optimization, Autotuning, Efficient ML Systems

Education

Seoul National University **Mar 2024 – Aug 2026**
Master of Data Science GPA 3.52/4.0
Computing Optimization and Data-driven Exploration Lab *Advisor: Hyojin Sung*
Thesis: Towards Batched Activation Sparsity in LLM Decoding

Yonsei University **Mar 2020 – Feb 2024**
B.A. in Applied Statistics GPA 3.92/4.3

University of California, Berkeley **Aug 2022 – May 2023**
UCEAP Exchange Program, College of Letters & Science

Research Experience

Computing Optimization and Data-driven Exploration (CODE) Lab – SNU **Sep 2024 – Present**
Graduate Researcher & Research Collaborator *Advisor: Hyojin Sung*

Autotuning Framework for High-Performance Computing Project

HYPERF (HPDC 2025)

- Evaluated HYPERF against OpenTuner on PolyBench kernels, demonstrating **6.0× average execution speedup** and 1.2× faster tuning.

Sparse Matrix-Vector Multiplication (SpMV) Optimization

- Profiled an SpMV optimization that groups matrix rows with similar numbers of nonzero elements to improve load balancing.
- Built an ML-based selector using XGBoost to predict when the optimization is beneficial and C5.0 to recommend tuning configurations.
- Integrated an ML-based prior into HYPERF v2's sampler, reaching within 5% of the best observed latency with **60% fewer trials** (570 → 230).

Autotuning Search and Cost Model Analysis (Ongoing)

- Investigating how cost-model predictions and random exploration affect TVM's search for high-performing programs.

Towards Batched Activation Sparsity in LLM Decoding (Master's thesis)

- Investigated whether blocked activation sparsity could be shared across sequences to accelerate batched LLM decoding without compromising model quality.
- Developed an oracle analysis and Triton block-sparse GEMM kernels to evaluate sparsity potential and GPU performance under batching.
- Found 64% average sparsity using independent per-step oracle analysis at batch size 16, corresponding to an **estimated 1.6× Triton block-sparse GEMM speedup** over dense cuBLAS.
- Identified the gap between oracle sparsity potential and practical online sparsification due to quality degradation and error accumulation during autoregressive decoding.

Publications

Juseong Park*, Yongwon Shin*, Junghyun Lee, **Junseo Lee**, Juyeon Kim, Oh-Kyoung Kwon, and Hyojin Sung. (*: co-first authors)
HYPERF: End-to-End Autotuning Framework for High-Performance Computing
In the 34th International Symposium on High-Performance Parallel and Distributed Computing (**HPDC**), 2025.

Relevant Coursework

Machine Learning for Systems *(Autotuning, Learned Scheduling & Code Optimization)*
Computing for Data Science 2 *(Parallel & GPU Computing)*
Big Data and Knowledge Management Systems 1 *(Database Systems)*
Programming Practice for Data Science *(Python/C++; Data Structures & Algorithms)*

Teaching & Mentoring

Undergraduate Data Science Project Tutor – SNU **Sep 2024 – Dec 2024**

- Mentored 30+ undergraduate students in Python programming and data science projects as part of a 4-person teaching team.

Skills

Programming: Python, C/C++, CUDA, Triton, Bash, R
Systems & Tools: TVM, NVIDIA Nsight Systems/Compute, OpenMP, Slurm, Docker, Git

Languages

Korean (native), English (professional), Spanish (basic)